Cell Biology & Biochemistry

Topics Covered
Biochemistry
Biomolecules and Molecular Organization
Biological Macromolecules:
- Carbohydrates
- Lipids
- Proteins
- Nucleic acids and Nucleotides
- Identification of Biomolecules (Practical)

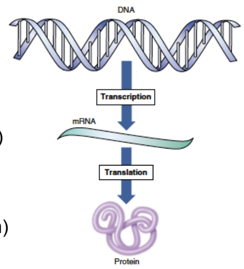
Cell Biology
Cell ultra-structure
Cell Division
Biological membranes: Structure and functions
Bioenergetics
Enzymes: General properties and mechanism of action
Cell signaling
Cellular respiration
DNA Replication
Protein Biosynthesis

Transcription and translation of genetic information

- Cells produce proteins by following the instructions stored in DNA
- This is done in two steps

  - Copying of the DNA sequence to RNA (mRNA) (**Transcription**)

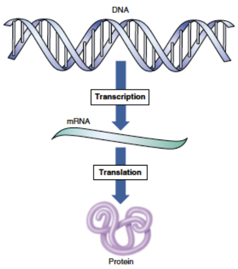
  - Formation of the proteins using amino acids according to the RNA sequence (**Translation**)



Transcription and translation of genetic information

Transcription: the production of messenger RNA (mRNA) by the enzyme RNA polymerase

Translation: use of mRNA to direct protein synthesis following the RNA sequence

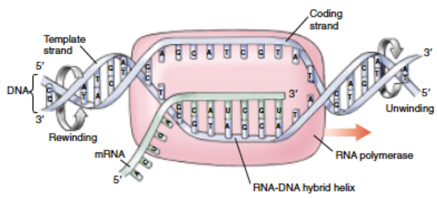


Transcription and translation of genetic information

Transcription
- mRNA copy of the DNA (i.e. gene) is copied (transcribed) from a DNA template
- Initiated when the enzyme **RNA polymerase** binds to the gene
- RNA polymerase moves along the DNA strand in the gene
- At each DNA nucleotide, it adds the corresponding complementary RNA nucleotide
- Four steps: Initiation, elongation, termination, post-transcriptional modification

Transcription and translation of genetic information

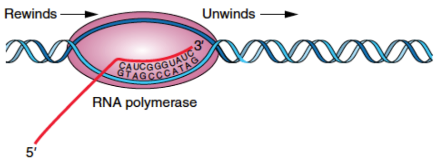
Transcription: Initiation
- RNA polymerase binds to the specific region of the DNA strand
- Unwinds and separates the DNA strand to form a small open complex (12-17 bp) – **transcription bubble**



Transcription and translation of genetic information

Transcription: Elongation

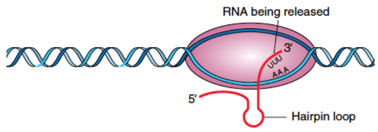
- RNA polymerase moves along the template strand, synthesizing an mRNA molecule
- Ribonucleotides are added in 5'-3' direction



Transcription and translation of genetic information

Transcription: Termination

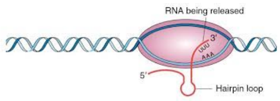
- At the end of the gene
  - Specific sequences causes the formation of phosphodiester bonds to cease
  - RNA-DNA hybrid within the transcription bubble dissociate
  - RNA polymerase release the DNA
  - DNA within the transcription bubble rewind



Transcription and translation of genetic information

Transcription: Termination

- At the end of the gene there is a sequence of nucleotides are rich in Guanine and Cytosine
- This creates a hairpin like structure – **GC hairpin**



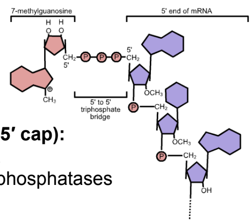
- The stress caused by the GC hairpin breaks the bonds in DNA-RNA hybrid
- mRNA transcript is then released from the ribosome

Transcription and translation of genetic information

Transcription: Post-transcriptional modification

- Eukaryotic mRNA transcripts are modified in several ways to aid the long journey out of the nucleus

- Addition of 5' caps
  - transcripts usually begin with A or G
  - the terminal **phosphate** of the 5' A or G is removed
  - 5'-5' linkage forms with GTP (**5' cap):** protects the 5' end of the RNA template from nucleases and phosphatases



Transcription and translation of genetic information

Transcription: Post-transcriptional modification

- Addition of 3' poly-A tails
  - The 3' end of transcript is cleaved off at a specific site containing the sequence AAUAAA
  - A special polymerase enzyme then adds about 250 Adenine ribonucleotides to the 3' end of the transcript: **Poly-A tail**
  - Poly-A tail protects the transcript from degradation by nucleases

Transcription and translation of genetic information

Translation

- mRNA transcript is used to direct the sequence of amino acids during the synthesis of polypeptide

- Nucleotide sequence of the mRNA transcript is translated into an amino acid sequence

- Involves 4 steps: Initiation, Elongation, Termination, Post-translational modification

- Translation takes place in the cytoplasm

Transcription and translation of genetic information

Translation

- Three nucleotides together in the mRNA strand determines the type of amino acid that is to be used in the protein - **codon**

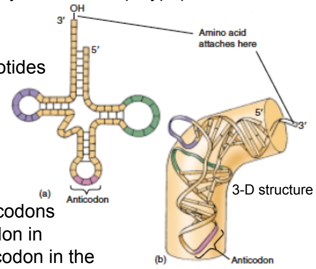
Second letter

First letter	U	C	A	G	Third letter
U	UUU Phe UUC Phe UUA Leu UUG Leu	UCU Ser UCC Ser UCA Ser UCG Ser	UAU Tyr UAC Tyr UAA Stop UAG Stop	UGU Cys UGC Cys UGA Stop UGG Trp	U C A G
C	CUU Leu CUC Leu CUA Leu CUG Leu	CCU Pro CCC Pro CCA Pro CCG Pro	CAU His CAC His CAA Gln CAG Gln	CGU Arg CGC Arg CGA Arg CGG Arg	U C A G
A	AUU Ile AUC Ile AUA Ile AUG Met	ACU Thr ACC Thr ACA Thr ACG Thr	AAU Asn AAC Asn AAA Lys AAG Lys	AGU Ser AGC Ser AGA Arg AGG Arg	U C A G
G	GUU Val GUC Val GUA Val GUG Val	GCU Ala GCC Ala GCA Ala GCG Ala	GAU Asp GAC Asp GAA Glu GAG Glu	GGU Gly GGC Gly GGA Gly GGG Gly	U C A G

Transcription and translation of genetic information

Translation

- tRNA molecules attached to Amino acids bring the Amino acids to the ribosome to synthesize the polypeptide chain
- tRNA molecules have a sequence of three nucleotides corresponding to a complementary codon in messenger RNA
 - **Anticodon**
- Specific tRNAs with anticodons corresponding to the codon in mRNA will pair with the codon in the mRNA strand

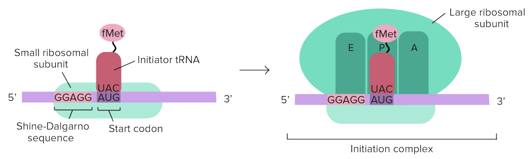


Transcription and translation of genetic information

Translation: Initiation

- Small subunit of the ribosome binds at the 5' end of the mRNA molecule
- Forms a complex with the large unit of the ribosome complex and an initiation tRNA molecule

Bacterial translation initiation



Transcription and translation of genetic information

Translation: Elongation

- Codons on the mRNA molecule determine which tRNA molecule linked to an amino acid binds to the mRNA
- The enzyme peptidyl transferase (carried out by the ribosome) links the amino acids together using peptide bonds
- The process continues, producing a chain of amino acids as the ribosome moves along the mRNA molecule

Transcription and translation of genetic information

Translation: Termination

- Translation is terminated when the ribosomal complex reaches the stop codon/s (UAA, UAG, UGA)
- Ribosomal complex in eukaryotes is larger and more complicated than in prokaryotes

Transcription and translation of genetic information

Translation: Post-translational modification

- Modifications occur on a polypeptide chain catalyzed by enzymes, after its translation by ribosomes
- Involves
 - folding of the polypeptide chain
 - addition of a functional group covalently to a protein

Transcription and translation of genetic information

Summary

- Cells produce proteins by following instructions stored in DNA
- This is done in two steps –Transcription and Translation
- Synthesis of mRNA is catalyzed by RNA polymerase enzyme
- GC hairpin in mRNA terminates transcription
- Translation takes place on ribosomes in the cytoplasm
- Three nucleotides together in the mRNA strand determine the type of amino acid that is to be used in the protein - codon
- tRNA molecules attached to Amino acids bring them to the ribosome to synthesize the polypeptide chain
- Peptidyl transferase enzyme links the amino acids together using peptide bonds producing a polypeptide chain
- Translation is terminated when the ribosomal complex reaches the stop codon